

At most d planks covering the sphere: total width at least two

Candidate partial result for arXiv:2211.10886

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Status. Candidate substantial partial result, likely valid, requiring human review. Conjecture 1.15 of Glazyrin–Karasev–Polyanskii is proved when the number of planks is at most the ambient dimension. More generally, it is proved whenever the plank normals do not span the ambient space.

1 Source conjecture

A (closed) real plank of Euclidean width w is a set

$$P = \{x \in \mathbb{R}^d : a \leq \langle u, x \rangle \leq b\}, \quad \|u\| = 1, \quad w = b - a.$$

Conjecture 1.15 of Glazyrin–Karasev–Polyanskii [1], originally Conjecture 1.17 of [2], asks whether, for $d \geq 4$, every finite plank cover of the unit sphere S^{d-1} has total width at least 2. The source statement is reproduced below.

We restate [10, Conjecture 1.17] here as an example of a question so far resisting the approach of Bang [6] (including its versions in [11, 17]) and the polynomial approach we currently consider.

Conjecture 1.15 (Conjecture 1.17 in [10]). *If $d \geq 4$ and the unit sphere $S^{d-1} \subset \mathbb{R}^d$ is covered by a finite number of real planks then the sum of their widths is at least 2.*

Figure 1: Conjecture 1.15 in the source PDF [1, p. 5].

2 Why the small-cardinality case becomes a ball problem

The complement of finitely many planks is a union of open polyhedral sign cells. If a point x is outside every plank, choose, for each plank, the one of its two outer open halfspaces containing x . Their intersection is the sign cell containing x .

With fewer than d independent normals, every such cell contains a line. With exactly d independent signed normals, solve one square linear system to find a direction that strictly increases all defining linear forms. Thus, for at most d planks, every nonempty sign cell has a ray to infinity. A ray starting inside the unit ball must cross its sphere. Consequently, covering the sphere already forces covering the whole ball, where Bang’s plank theorem applies.

3 The partial theorem

Theorem 1. *Let $P_1, \dots, P_N$ be real planks covering $S^{d-1} \subset \mathbb{R}^d$, and let u_i be a unit normal of P_i . If either*

1. $N \leq d$, or
2. $\text{span}\{u_1, \dots, u_N\} \neq \mathbb{R}^d$,

then the same planks cover the closed unit ball B^d . In particular,

$$\sum_{i=1}^N \text{width}(P_i) \geq 2.$$

Proof. Write

$$P_i = \{y \in \mathbb{R}^d : a_i \leq \langle u_i, y \rangle \leq b_i\}.$$

Suppose, for a contradiction, that some $x \in B^d$ is outside every P_i . Because the sphere is covered, $\|x\| < 1$. For each i , exactly one of the strict inequalities

$$\langle u_i, x \rangle > b_i, \quad \langle -u_i, x \rangle > -a_i$$

holds. Choose $v_i \in \{u_i, -u_i\}$ and $c_i \in \{b_i, -a_i\}$ so that $\langle v_i, x \rangle > c_i$. The open sign cell

$$C = \{y \in \mathbb{R}^d : \langle v_i, y \rangle > c_i \text{ for every } i\}$$

therefore contains x and is disjoint from every plank.

First suppose the normals span a proper subspace. Choose a nonzero vector v orthogonal to all v_i . Then $x + tv \in C$ for every $t \geq 0$.

It remains to treat the first hypothesis when the normals span $\mathbb{R}^d$. Since $N \leq d$, necessarily $N = d$, and $v_1, \dots, v_d$ are linearly independent. Hence the square system

$$\langle v_i, v \rangle = 1 \quad (1 \leq i \leq d)$$

has a solution v . For every $t \geq 0$,

$$\langle v_i, x + tv \rangle = \langle v_i, x \rangle + t > c_i,$$

so again $x + tv \in C$.

In either case C contains a ray $x + tv$ with $v \neq 0$. Its norm starts below 1 and tends to infinity. By continuity the ray meets S^{d-1} , at a point still in C and hence outside every plank. This contradicts sphere coverage. Thus the planks cover B^d .

Bang's plank theorem [3] says that planks covering a convex body have total width at least its minimal width. The unit ball has minimal width 2, which proves the final assertion. $\square$

Corollary 2. *In $\mathbb{R}^4$, Conjecture 1.15 holds for every cover of S^3 by at most four planks. More generally, in every dimension it holds for at most d planks. Any counterexample must use at least $d + 1$ planks whose normals span $\mathbb{R}^d$.*

Proof. The positive claims are Theorem 1. Its contrapositive gives the two necessary conditions on a counterexample. $\square$

4 Scope, obstruction, and verification

The argument does not settle the full conjecture. Already $d + 1$ suitably oriented open halfspaces can have a bounded simplex as their intersection. Equivalently, the selected signed normals can positively span $\mathbb{R}^d$, so the common recession direction used above need not exist. A direct attempt to compare the facet distances of such a bounded uncovered cell with the widths of the original planks does not close: the opposite boundary of each plank generally cuts a different neighboring cell. The two polynomial-plank conjectures in the source are also unaffected.

No computation enters the proof. The only external theorem is Bang’s classical plank theorem. Strict inequalities are preserved along the ray, so there is no boundary or closure issue.

Bounded novelty check. The original conjecture paper records the $d = 3$ case and, for $d \geq 4$, a reduction when there are at most three planks [2, Remark after Conjecture 1.17]. A current survey still lists the conjecture as open and records the same S^3 at-most-three-plank subcase [4]. The run’s registry, solution, attempt, and proof-gap indexes, plus bounded exact-phrase, boundary-covering, four-plank, cardinality, title, DOI, and citing-work searches through 11 August 2026, found no statement of the $N \leq d$ theorem or the proper-normal-span extension. Novelty confidence is moderate; this short polyhedral observation could be folklore, and no priority claim is made.

Human-review recommendation. Check the sign-cell construction, the square system in the full-rank $N = d$ case, and the ray–sphere crossing. Then independently search the classical spherical-plank literature for this cardinality reduction. Subject to that audit, classify the result as a likely valid substantial partial advance, not as a full solution.

References

- [1] A. Glazyrin, R. Karasev, and A. Polyanskii, *Extensions of polynomial plank covering theorems*, Bull. Lond. Math. Soc. **56** (2024), 1014–1028; arXiv:2211.10886.
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- [3] T. Bang, *A solution of the “plank problem”*, Proc. Amer. Math. Soc. **2** (1951), 990–993.
- [4] W. Verreault, *Plank theorems and their applications: a survey*, Bull. Lond. Math. Soc. (2026), DOI: 10.1112/blms.70230; arXiv:2203.05540v2.